

- 5 (a) State what is meant by the term *internal energy of an ideal gas*.

The internal energy of an ideal gas is the sum of the microscopic kinetic energy due to random motion of all the atoms/molecules of the gas.

- (b) The pressure p exerted by an ideal gas is given by the equation

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

State what the symbols ρ and $\langle c^2 \rangle$ represent.

ρ : density of the gas

$\langle c^2 \rangle$: mean-square speed of the atoms/molecules in the gas

- (c) Use the equation given in (b) to show that the total kinetic energy of the molecules of an ideal gas of volume V and pressure p is given by $\frac{3}{2} pV$.

$$p = \frac{1}{3} \rho \langle c^2 \rangle = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

where N is the number of molecules in the gas; m the mass of each molecule and V the volume of the gas. Therefore,

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

$$\frac{1}{2} Nm \langle c^2 \rangle = \frac{3}{2} pV$$

- (d) (i) Calculate the internal energy of an ideal gas of volume $3.4 \times 10^{-4} \text{ m}^3$ when its pressure is 100 kPa.

$$\begin{aligned}\text{Internal Energy} &= \frac{3}{2}pV \\ &= \frac{3}{2}(100 \times 10^3)(3.4 \times 10^{-4}) \\ &= 51 \text{ J}\end{aligned}$$

- (ii) Calculate the increase in internal energy when the gas expands at constant pressure to a volume of $8.8 \times 10^{-4} \text{ m}^3$.

$$\begin{aligned}\text{Increase in internal Energy} &= \frac{3}{2}p\Delta V \\ &= \frac{3}{2}(100 \times 10^3)(8.8 \times 10^{-4} - 3.4 \times 10^{-4}) \\ &= 81 \text{ J}\end{aligned}$$

- (iii) Determine the heat supplied to the gas so as to enable the expansion at constant pressure to take place.

$$\begin{aligned}\text{Work done by gas} &= p\Delta V \\ &= (100 \times 10^3)(8.8 \times 10^{-4} - 3.4 \times 10^{-4}) \\ &= 54 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta U &= Q + W \\ 81 &= Q - 54 \\ Q &= 135 \text{ J}\end{aligned}$$